Retail Sales Data Analysis & Business Intelligence Dashboard

Role: Data Analyst (Project)

Tools Used: Python (Pandas), Google Colab, Power BI, Excel

1. Business Problem

Retail businesses need to understand sales performance, product demand, and regional profitability to improve decision-making. This project analyzes retail sales data to identify key trends and build a dashboard that helps stakeholders monitor business performance.

2. Objectives

- Analyze sales performance across product categories
- Identify the most profitable regions and products
- Understand monthly sales trends
- Build an interactive Power BI dashboard for business insights

3. Dataset Information

The dataset contains retail sales transaction data with the following attributes:

Column	Description
Order Date	Transaction date
Product Name	Name of the product sold
Category	Product category
Region	Sales region
Quantity	Units sold
Sales	Revenue generated
Cost	Cost of goods sold
Profit	Profit earned

4. Data Cleaning (Python)

Data cleaning was performed using Python (Pandas) in Google Colab. Steps included checking missing values, verifying data types, and converting the Order Date column into datetime format.

Example Python code:

```
data.isnull().sum()

data['Order Date'] = pd.to_datetime(data['Order Date'])
```

5. Exploratory Data Analysis (EDA)

Key analyses performed:

- Sales by Category
- Profit by Region
- Top 10 Products by Sales
- Monthly Sales Trend Analysis

6. Data Visualization

Visualizations were created using Python (Matplotlib) and Power BI to understand sales performance and identify key patterns.

Charts created:

- Sales by Category
- Profit by Region
- Top 10 Products by Sales
- Monthly Sales Trend

7. Power BI Dashboard

An interactive Power BI dashboard was developed to present insights visually. The dashboard includes KPI cards, charts, and filters for interactive analysis.

Dashboard Components:

- KPI Cards: Total Sales, Total Profit, Total Quantity Sold

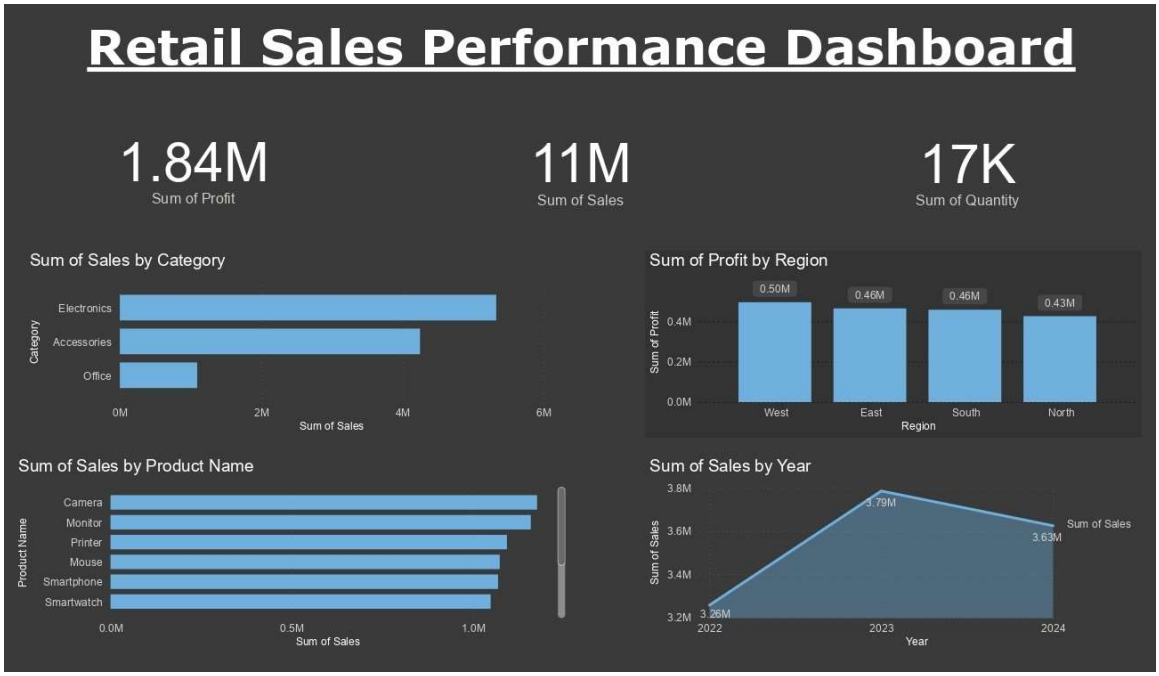
• Sales by Category chart

• Profit by Region chart

• Top 10 Products by Sales chart

• Monthly Sales Trend line chart

• Interactive filters (Region, Category, Order Date)



8. Key Insights

• Certain product categories generate significantly higher revenue.

• Some regions contribute more profit than others.

• A small number of products account for a large share of total sales.

• Sales fluctuate across months indicating seasonal demand.

9. Business Recommendations

• Focus marketing efforts on high-performing products.

• Improve strategies in lower-performing regions.

• Maintain inventory for top-selling products.

• Use sales trends for demand forecasting.

10. Skills Demonstrated

• Data Cleaning and Preparation

• Exploratory Data Analysis (EDA)

• Python (Pandas)

• Data Visualization

• Power BI Dashboard Development

• Business Insights and Reporting

11. Resume Project Entry

Retail Sales Data Analysis & Dashboard | Python, Power BI
• Analyzed retail sales dataset using Python (Pandas) to identify revenue and profit trends.
• Performed data cleaning, validation, and exploratory data analysis.
• Developed an interactive Power BI dashboard with KPI metrics and dynamic filters.
• Generated business insights on category performance, regional profitability, and top products.